

NAME

mbimagelist – parses recursive imagelist structures, performing one or more tasks on the image files referenced in the imagelist(s). The default action is to print out the complete list of image file paths; other possible tasks include also printing out associated timestamps and camera settings, and printing out processing parameter values embedded in the imagelist structure.

VERSION

Version 5.0

SYNOPSIS

```
mbimagelist
[
  --absolutepaths {-A}
  --copy=directory {-Cdirectory}
  --copyhere
  --files {-F}
  --help {-H}
  --input=file {-Ifile}
  --left {-L}
  --parameters {-P}
  --right {-R}
  --settings {-S}
  --single
  --verbose {-V}
]
```

DESCRIPTION

MBimagelist is a utility for parsing imagelist files. Imagelist files, or lists of seafloor photographic images, are used by a number of **MB-System** programs dealing with seafloor photography. These lists may contain references to other imagelists, making them recursive. By default, the program **mbimagelist** outputs each image filename encountered as it parses through the input imagelist tree. The output paths can be relative to the current working directory or absolute.

Imagelist files can also contain processing parameters associated with the images, including specifying a file containing the calibration parameters for the camera rig, specifying a file containing the trajectory (navigation and attitude) of the camera rig as images were collected, specifying a file with an image correction lookup table, and several other parameters. All of these parameters correspond to command line options for **mbphotomosaic** that can also be specified.

This program can be used in shellscripts to read imagelists in the same fashion as **MB-System** programs like **mbphotomosaic** and **mbphotocorrect**. This program can also be used to check and debug complex recursive imagelist structures.

MB-SYSTEM AUTHORSHIP

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OPTIONS

--absolutepaths {-A}

Causes the image file paths in the output to be converted to absolute paths (by prepending the current working directory to any relative path) rather than being output as they appear in the imagelist. Default: paths are output exactly as they appear in the imagelist.

--copy=directory {-C}directory}

Causes each image file listed in the output to also be copied (using the shell **cp** command) into *directory*.

--copyhere

Equivalent to **--copy**, except that the destination directory is the current working directory rather than one specified by the user.

--files {-F}

Requests that the plain image file paths be listed, one per line. This is the default output mode when neither **--parameters** nor **--settings** is specified.

--help {-H}

This "help" flag cause the program to print out a description of its operation and then exit immediately.

--input=file {-I}file}

Sets the filename of the imagelist to be parsed. Imagelists are recursive, and may reference other imagelists as well as individual image files. Default: *file* = "imagelist.mb-2".

--left {-L}

Sets the image selection to left-camera images. This option is reserved for future use; the current release of **mbimagelist** lists every image referenced in the imagelist regardless of this setting.

--parameters {-P}

Causes a line to be printed for each processing parameter entry encountered while parsing the imagelist, in addition to (or instead of, if **--files** and **--settings** are not also given) the image file listing. Default: parameter entries are not reported.

--right {-R}

Sets the image selection to right-camera images. This option is reserved for future use; the current release of **mbimagelist** lists every image referenced in the imagelist regardless of this setting.

--settings {-S}

Causes each image line in the output to include the image timestamp, gain, and exposure values, in addition to the file path. Default: only the file path is printed.

--single

Sets the image selection to single-camera (non-stereo) images. This option is reserved for future use; the current release of **mbimagelist** lists every image referenced in the imagelist regardless of this setting. There is no short option equivalent.

--verbose {-V}

Increases the verbosity of **mbimagelist**. Can be used multiple times. If used at all, a summary of the numbers of left, right, single, and total images listed is printed at the end of execution.

EXAMPLES

Suppose we have two swath data files from an EM3000 multibeam and another two from an Hydrosweep MD multibeam. We might construct two imagelist files. For the EM3000 we might have a file *imagelist_em3000.mb-1* containing:

```
0004_20010705_165004_raw.mb57 57
0005_20010705_172010_raw.mb57 57
```

For the Hydrosweep MD data we might have a file imagelist_hsmc.mb-1 containing:

```
al10107051649.mb102 102
al10107051719.mb102 102
```

Further suppose that we have found it necessary to edit the bathymetry in 0005_20010705_172010_raw.mb57 and al10107051719.mb102 using **mbedit**, and that **mbprocess** has been run on both files to generate processed files called 0005_20010705_172010_rawp.mb57 and al10107051719p.mb102.

If we run:

```
mbimagelist --input=imagelist_em3000.mb-1
```

the output is:

```
0004_20010705_165004_raw.mb57 57 1.000000
0005_20010705_172010_raw.mb57 57 1.000000
```

Here the file name is followed by the format and then by a third column containing the default file weight of 1.0.

Similarly, if we run:

```
mbimagelist --input=imagelist_hsmc.mb-1
```

the output is:

```
al10107051649.mb102 102 1.000000
al10107051719p.mb102 102 1.000000
```

If we insert a line

```
$PROCESSED
```

at the top of both imagelist_hsmc.mb-1 and imagelist_em3000.mb-1, then the output of **mbimagelist** changes so that:

```
mbimagelist --input=imagelist_em3000.mb-1
```

yields:

```
0004_20010705_165004_raw.mb57 57 1.000000
0005_20010705_172010_rawp.mb57 57 1.000000 and:
mbimagelist --input=imagelist_hsmc.mb-1
```

yields:

```
al10107051649.mb102 102 1.000000
al10107051719p.mb102 102 1.000000
```

Now suppose we create a imagelist file called imagelist_all.mb-1 that refers to the two imagelists shown above (without the \$PROCESSED tags). If the contents of imagelist_all.mb-1 are:

```
imagelist_em3000.mb-1 -1 100.0
imagelist_hsmc.mb-1 -1 1.0
```

where we have specified different file weights for the two imagelists, then:

```
mbimagelist --input=imagelist_all.mb-1
```

yields:

```
0004_20010705_165004_raw.mb57 57 100.000000
0005_20010705_172010_raw.mb57 57 100.000000
al10107051649.mb102 102 1.000000
al10107051719p.mb102 102 1.000000
```

SEE ALSO

mbsystem(1)

BUGS

No true bugs here, only distantly related arthropods... Yum. Seriously, it would be better if the copy function preserved the modification times of the copied swath files and ancillary files. Copying of processed files should also be an option.